

CUSTOMER CHURN

Exploratory Data Analysis Report

Identifying behavioural, account and service factors associated with customer attrition

1,000,000 customer records	31 cleaned variables	9.9% overall churn rate
26.5% month-to-month churn	22.1% churn with 4+ complaints	17.2% churn with 3+ late payments

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Portfolio report based on exploratory analysis of a cleaned synthetic telecommunications customer dataset.

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How to read this report

The full Jupyter notebook contains all code, distributions and intermediate summaries. This report focuses on the findings most useful for communicating the project to recruiters and business stakeholders.

Executive Summary

This report examines which customer characteristics, account structures, service experiences and payment behaviours are most strongly associated with churn in a cleaned synthetic telecommunications dataset containing 1,000,000 customer records. The analysis was designed to convert a large set of descriptive results into practical retention insights while clearly separating strong patterns from variables with little standalone value.

The clearest findings were concentrated in behavioural and account-related factors. Month-to-month customers recorded a churn rate of 26.5%, compared with 12.7% for one-year customers and 5.7% for two-year customers. Churn also increased consistently with complaint volume, rising from 8.1% among customers with no complaints to 22.1% among customers with four or more complaints. Service-call frequency showed a similar escalation, reaching 19.7% at six calls and 31.2% at ten calls, although the highest call-count categories contained relatively few customers.

Customer satisfaction, late payments and service adoption provided further actionable signals. Churn fell from 16.7% at the lowest satisfaction score to 7.5% at the highest. It rose from 9.1% for customers with no late payments to 17.2% for those with three or more. Customers subscribed to six services churned at 7.4%, compared with 11.0% among customers with only one service. Technical support and online security were the individual services most clearly associated with lower churn.

In contrast, age, gender, education, marital status, annual income, payment method, paperless billing, credit score, days since last interaction and average monthly data usage showed little meaningful standalone relationship with churn. The overall implication is that retention strategies should prioritise observable customer experience and account signals rather than demographic targeting.

Primary conclusion

The most useful EDA-stage warning signals are complaint history, service-call frequency, contract type, low satisfaction and repeated late payments. These associations should be validated on a consistent scale during predictive modelling before they are treated as formal decision rules.

1. Business Context and Objectives

Customer churn is costly because the loss of an existing customer can reduce recurring revenue and increase the need for acquisition spending. A useful churn analysis therefore needs to do more than describe who leaves; it should identify observable signals that could support earlier and more targeted intervention.

The exploratory analysis was organised around eight business questions covering customer segments, lifecycle stage, service experience, disengagement, value, service adoption, payment behaviour and the relative strength of different factor groups. The main objectives were to:

- Identify customer groups with meaningfully different churn rates.
- Distinguish actionable behavioural and account signals from weak demographic patterns.
- Engineer and evaluate a simple value-related feature: monthly charge per subscribed service.
- Generate evidence-based retention hypotheses for later machine-learning validation.

Scope

This report covers exploratory analysis only. The results describe associations observed in the dataset and do not establish causation or final predictive importance.

2. Dataset and Analytical Approach

The cleaned dataset contains 1,000,000 customer records and 31 variables spanning demographics, contracts, tenure, service subscriptions, usage, customer satisfaction, complaints, service contacts, billing, credit and churn status. The target is imbalanced: approximately 90.1% of customers were retained and 9.9% churned.

Area	Approach
Data quality	Unique customer records, no duplicates, completed missing-value treatment and consistent category labels.
Categorical comparisons	Churn rates compared across contract, demographic, service and payment groups.
Numerical comparisons	Means, medians, normalized distributions, boxplots and Pearson correlations.
Count variables	Grouped churn rates for complaints, service calls and late payments, with sample sizes reviewed.
Feature engineering	Monthly charges divided by number of services to create monthly charge per service.
Interpretation	Practical differences prioritised over statistical significance, given the very large sample size.

Average monthly data usage originally contained approximately 5% missing values. Because usage depends strongly on whether a customer has internet service, missing values were imputed using group-based medians by internet-service status rather than one overall median. The analysis also noted potential extreme values in monthly charges and treated the corresponding visual comparisons cautiously.

3. Exploratory Findings

3.1 Contract Structure and Customer Lifecycle

Contract type produced the largest account-level difference in churn. Month-to-month customers churned at 26.5%, more than twice the rate for one-year contracts and nearly five times the rate for two-year contracts. This does not prove that a longer contract independently prevents churn, but it identifies month-to-month customers as a high-priority segment for modelling and retention analysis.

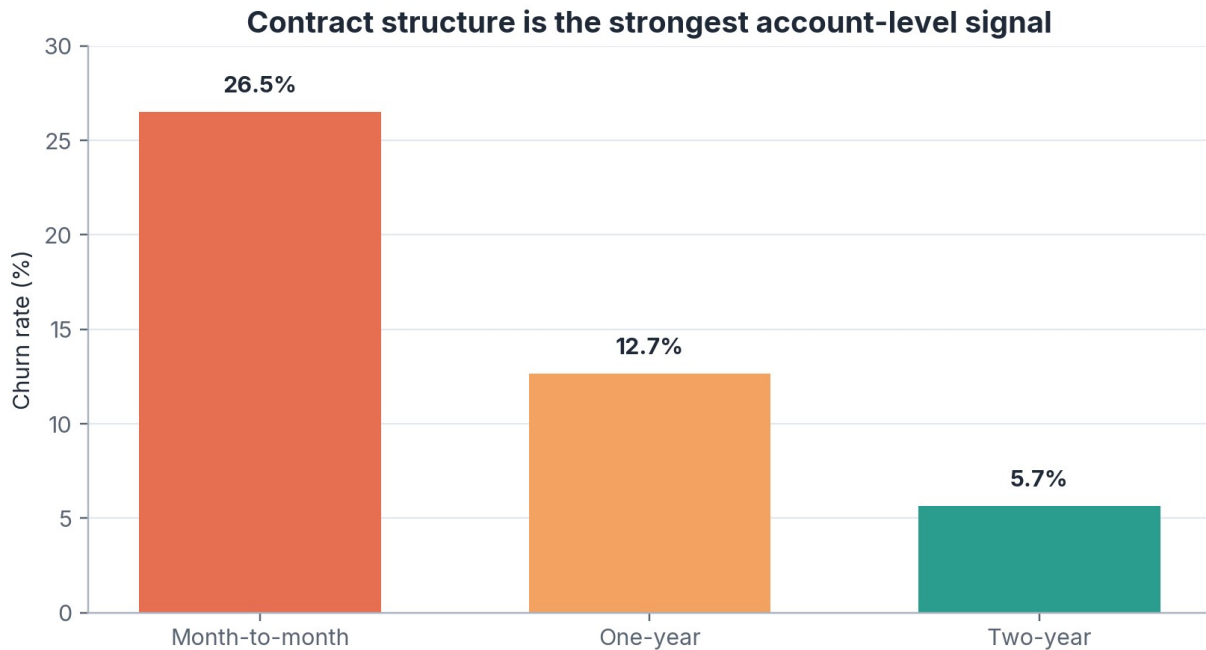


Figure 1. Churn rate by contract type. Source: author analysis.

Tenure showed a smaller but consistent lifecycle pattern. Churn was highest during the first six months at 10.39% and generally declined toward approximately 9.2% after four years. Churned customers also had a slightly shorter mean tenure (21.6 months) than retained customers (22.5 months). The difference is useful as context, but tenure alone is unlikely to distinguish high-risk customers effectively.

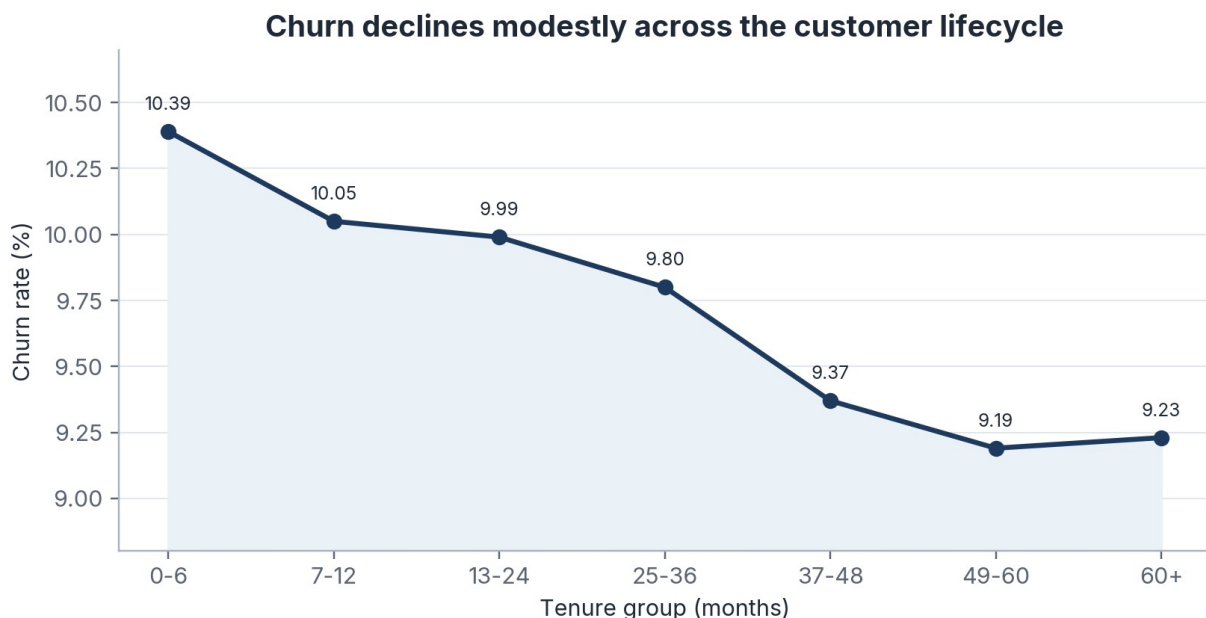


Figure 2. Churn rate by tenure group. Source: author analysis.

3.2 Demographic Characteristics

Demographic variables showed almost no practical separation. Churn rates across age groups ranged from 9.90% to 9.99%; gender, education and marital-status rates were similarly concentrated around the overall 9.9% churn rate. Retained and churned customers had almost identical annual incomes, including the same median of approximately \$48,955.

Variable	Observed churn-rate range / comparison	Interpretation
Age group	9.90%-9.99%	Negligible variation
Gender	9.82%-9.93%	Negligible variation
Education	9.86%-10.00%	Negligible variation
Marital status	9.70%-9.98%	Small variation
Annual income	\$58,488 retained vs \$58,545 churned mean	No meaningful difference

Business implication

Demographic-based targeting would add little value as a standalone retention strategy in this dataset. Customer experience and account behaviour provide substantially clearer signals.

3.3 Customer Satisfaction and Service Friction

Service experience produced some of the clearest and most actionable patterns in the EDA. Churn declined steadily as satisfaction increased, with a particularly noticeable reduction between satisfaction scores of three and four. Customers scoring three or below therefore represent a plausible warning segment for further modelling.

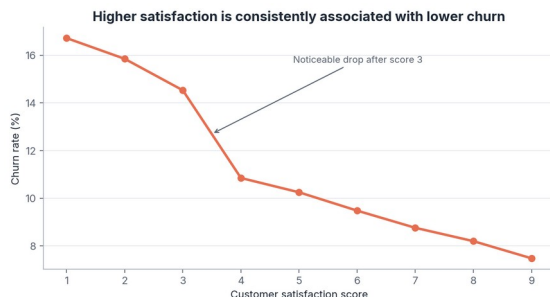


Figure 3. Churn rate by satisfaction score.



Figure 4. Churn rate by complaint count.

Complaint history showed a strong escalation pattern: churn increased from 8.1% with no complaints to 10.4% after one, 13.4% after two, 17.7% after three and 22.1% with four or more. The relationship is both large and consistent, making complaint volume a strong candidate for proactive service recovery.

Service-call frequency showed a similar pattern. Churn rose from 7.7% among customers with no service calls to 15.0% at five calls and 19.7% at six calls. It continued upward to 31.2% at ten calls. The apparent decline at eleven and twelve calls should not be treated as a genuine reversal because those categories contained only 68 and 36 customers respectively.

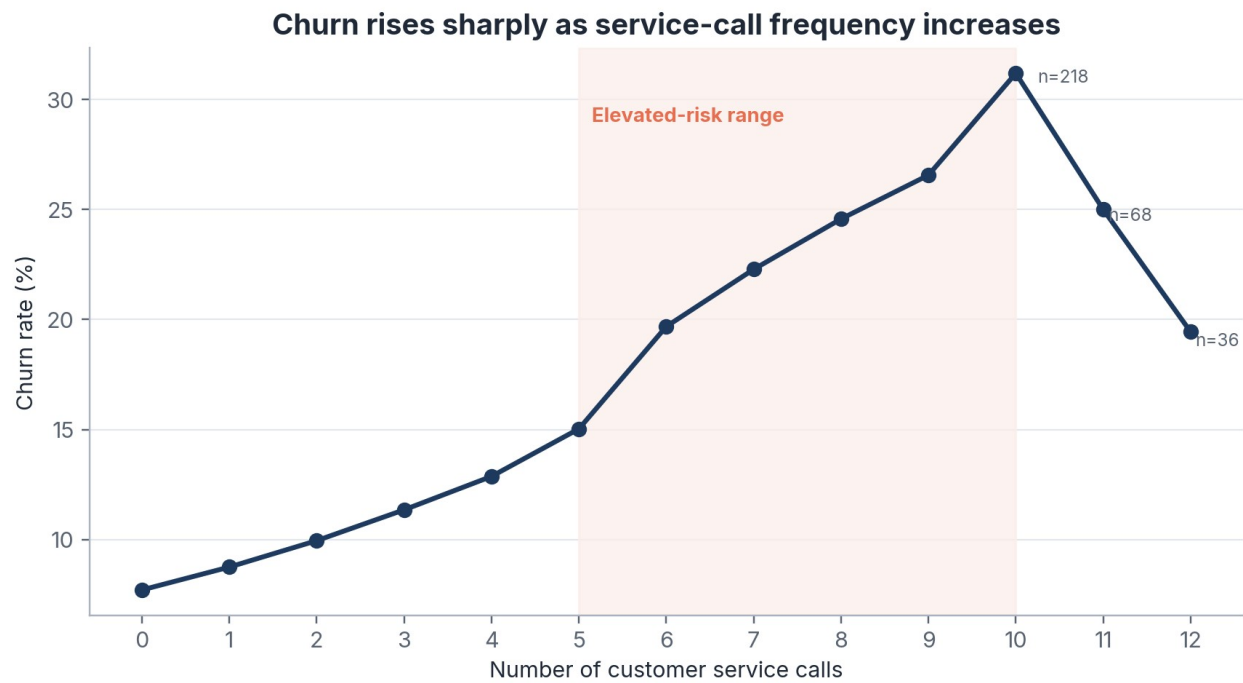


Figure 5. Churn rate by number of service calls. High-count categories should be interpreted cautiously because of small samples.

Potential operational trigger

A combination of low satisfaction, repeated complaints and approximately five or more service calls could be tested as a service-friction alert during modelling. The final threshold should be selected using validation data and business costs, not EDA alone.

3.4 Engagement, Pricing and Customer Value

The two direct disengagement measures were weak. Churned and retained customers both had a median of 31 days since their last interaction and almost identical means of 44.55 and 44.49 days. Average monthly usage also differed only slightly: retained customers averaged 38.64 GB, compared with 37.75 GB for churned customers. A sampled joint analysis did not reveal a distinct high-risk cluster.

Raw price measures were similarly limited. Churned customers paid slightly less per month on average (\$85.29 versus \$86.57), and their lower total charges were consistent with their slightly shorter tenure rather than an independent pricing effect.

The engineered monthly charge per service measure produced a clearer pattern. Churned customers paid an average of \$45.40 per service compared with \$43.42 for retained customers. Across deciles, churn increased from 8.61% among customers paying the least per service to 11.27% among those paying the most. Because this measure is partly related to the number of subscribed services, it should be treated as a value proxy and tested alongside its component variables in modelling.

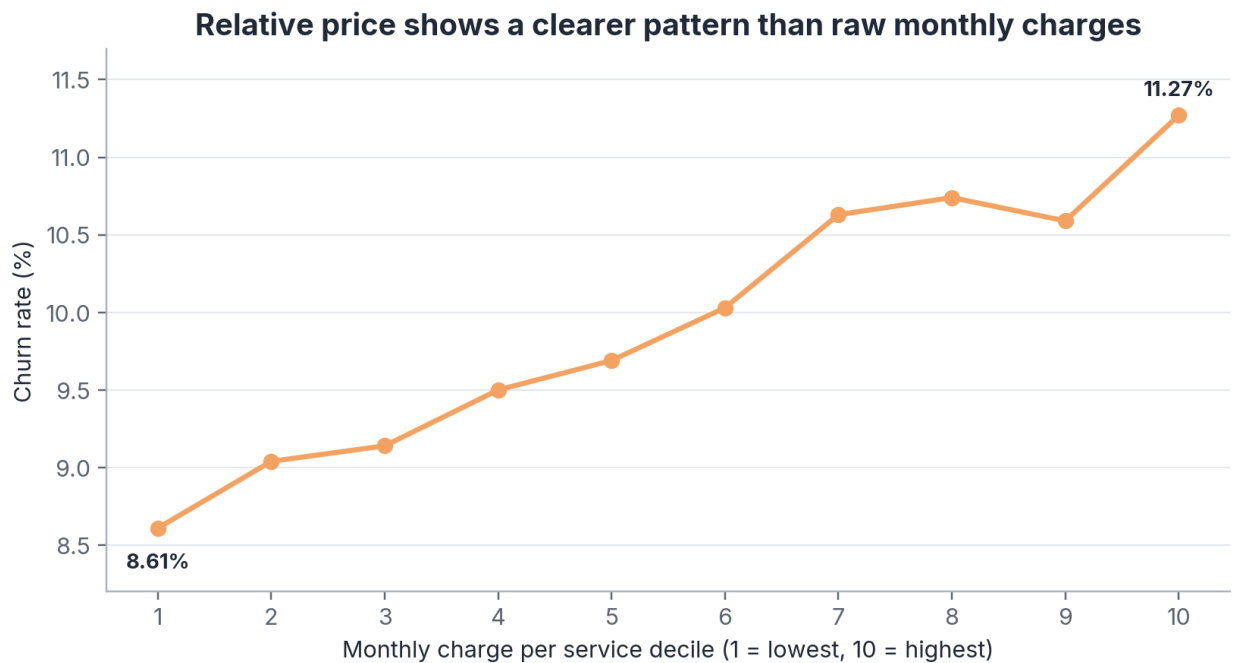


Figure 6. Churn rate across monthly charge per service deciles. Source: author analysis.

3.5 Service Adoption and Retention

Not all services were equally associated with retention. Technical support and online security showed the largest differences: customers with technical support churned at 8.56% compared with 11.28% among those without it, while online-security customers churned at 8.50% compared with 10.65%. Entertainment add-ons showed almost no difference.

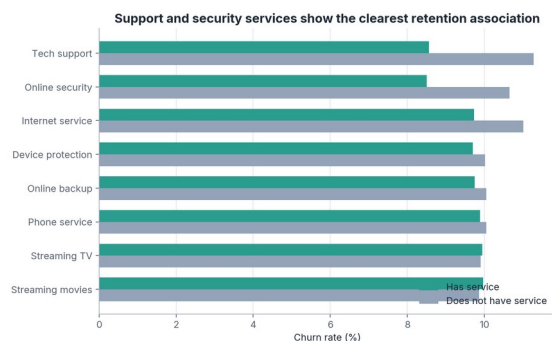


Figure 7. Churn rates with and without individual services.

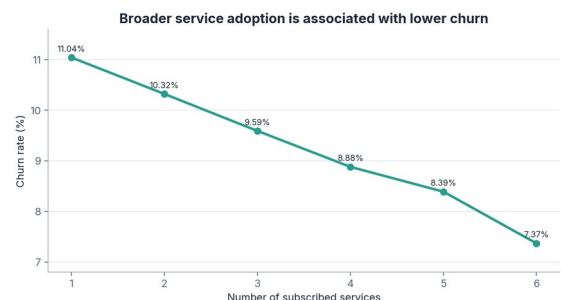


Figure 8. Churn rate by number of services.

The total number of subscribed services produced a clean monotonic trend. Churn fell from 11.04% among customers with one service to 7.37% among customers with six. This suggests that broader service adoption is a useful retention-related signal, although service mix, contract type and customer tenure may partly explain the relationship.

3.6 Payment and Billing Behaviour

Payment method, paperless billing and credit score were weak indicators. Churn rates across payment methods ranged only from 9.88% to 9.95%; paperless billing differed by 0.09 percentage points; and churned and retained customers had nearly identical mean credit scores (678.52 versus 678.63) and identical medians of 680.

Late payments were the clear exception. Churn increased steadily from 9.06% with no late payments to 11.13% with one, 13.60% with two and 17.16% with three or more. This makes late-payment history one of the most observable and operationally useful financial warning signals identified during the EDA.

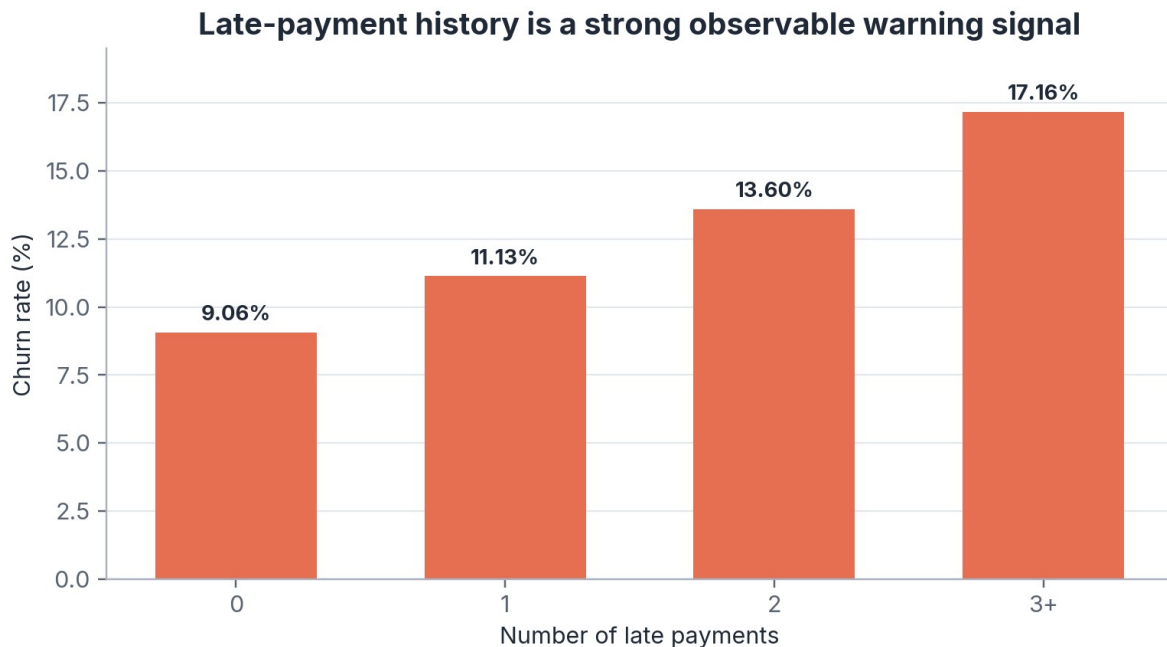


Figure 9. Churn rate by late-payment group. Source: author analysis.

4. Cross-Factor Synthesis

To compare grouped and categorical factors on a common descriptive basis, the analysis calculated the difference between the highest and lowest churn rates within each variable. Service calls, contract type and grouped complaints produced spreads of approximately 14-23 percentage points, followed by satisfaction and late payments. All demographic variables remained below 0.3 percentage points.

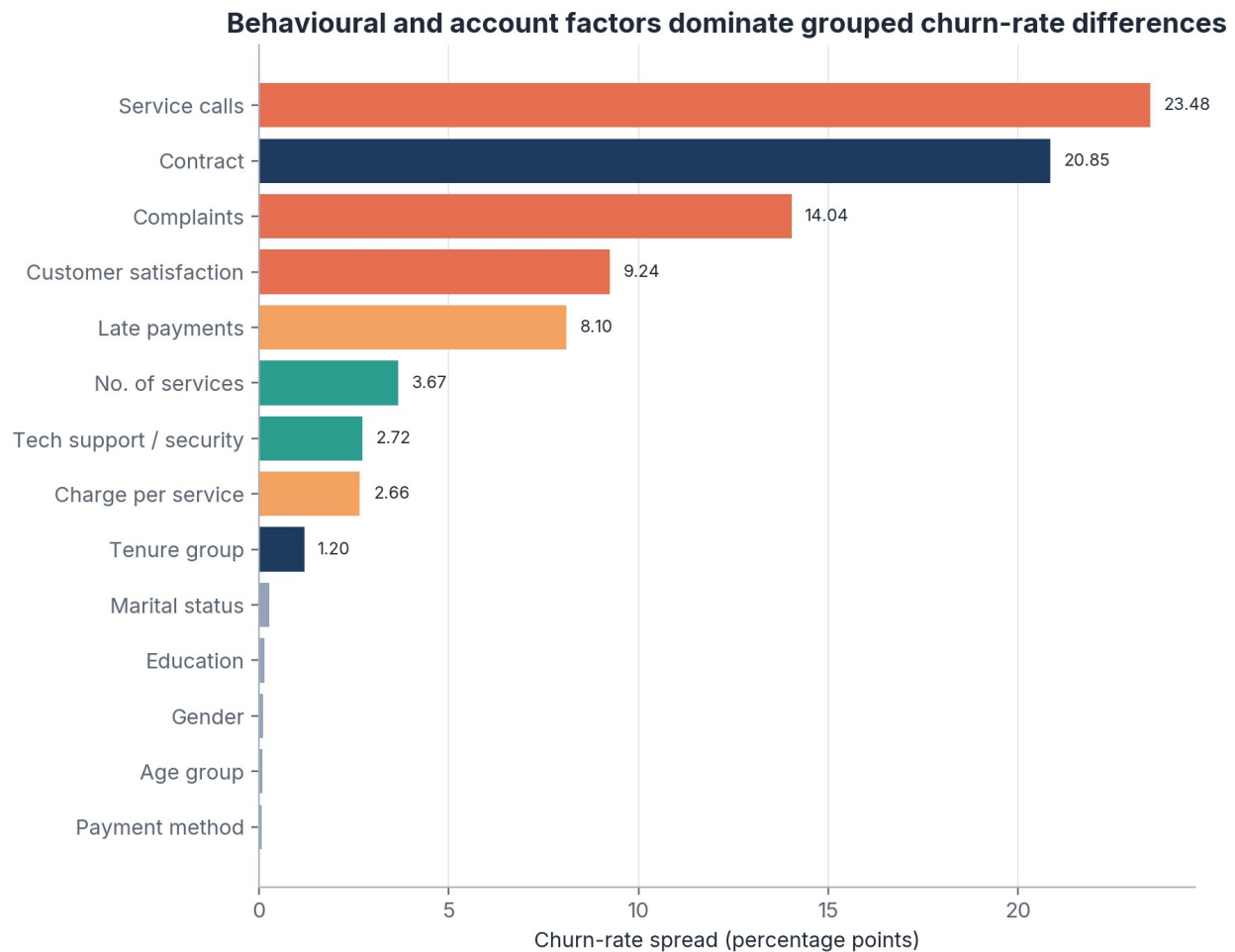


Figure 10. Churn-rate spread across grouped and categorical variables. Percentage-point spreads are not directly comparable with correlation coefficients.

A separate correlation review was used for continuous variables not otherwise represented through grouped analysis. Average monthly usage had an absolute correlation with churn of approximately 0.02, while annual income, days since last interaction and credit score were all approximately 0.01 or lower. These coefficients were kept separate from percentage-point spreads because the two measures use different scales.

Synthesis

The qualitative result is consistent across the EDA: behavioural and account information carries substantially more standalone churn signal than demographics. The strongest candidates for modelling are complaint volume, service-call frequency, contract type, satisfaction, late payments, service count and monthly charge per service.

5. Business Recommendations

1. Build a service-friction monitoring layer

Track complaint count, service-call frequency and satisfaction together. Customers with repeated complaints, low satisfaction or roughly five or more calls should be prioritised for service recovery and reviewed during model threshold design.

2. Prioritise month-to-month customers

Month-to-month customers form the clearest high-churn account segment. Test targeted onboarding, loyalty offers and voluntary movement toward longer commitments, while controlling for tenure and service mix.

3. Use late payments as an early warning signal

Repeated late payments are easy to observe and show a strong monotonic relationship with churn. Combine payment history with service-friction signals rather than treating it as a stand-alone credit judgement.

4. Strengthen support-oriented bundles

Technical support and online security are associated with lower churn, and broader service adoption is linked to stronger retention. Test whether support and security bundles increase engagement without relying on entertainment add-ons.

5. Avoid demographic-first targeting

Age, gender, education, marital status and income showed little practical value. Retention resources should focus on customer experience and account behaviour, which are more actionable and less likely to produce unhelpful demographic segmentation.

6. Validate all interventions through modelling and experimentation

Use train/validation/test splits, class-sensitive metrics and model interpretation to confirm whether the EDA signals remain useful jointly. Any customer-facing intervention should then be evaluated through controlled testing.

6. Limitations and Next Steps

- The dataset is synthetic, so patterns may not transfer directly to a real telecommunications customer base.
- Most EDA comparisons are univariate and may reflect confounding between contract, tenure, service count and service mix.
- The analysis identifies association rather than causation; interventions may not reproduce the observed differences.
- Very high complaint and service-call categories contain small samples and should be interpreted cautiously.
- Average monthly usage includes group-based median imputation, which creates structural concentrations in the distribution.
- Extreme monthly-charge and charge-per-service values require additional review during preprocessing.
- Individual internet add-on comparisons may mix customers without internet service with internet customers who declined the add-on, so conditional analysis is needed.
- The 9.9% churn rate creates class imbalance; accuracy alone will be insufficient for model evaluation.

The next stage should build a reproducible preprocessing and modelling pipeline using logistic regression as an interpretable baseline, followed by tree-based models such as random forest and XGBoost. Evaluation should include precision, recall, F1-score, ROC-AUC and preferably precision-recall AUC, with threshold selection guided by the cost of missed churners and unnecessary interventions. Feature importance or SHAP analysis can then determine whether the EDA findings remain important when variables are considered jointly.

Appendix A. Selected Statistical Summary

The tables below collect the main figures referenced in the report. Values are descriptive and rounded to two decimal places.

A1. Strongest grouped churn-rate patterns

Factor	Low-risk / reference group	Higher-risk group	Spread
Complaints	0: 8.10%	4+: 22.14%	14.04 pp
Service calls	0: 7.71%	10: 31.19%	23.48 pp
Contract	Two-year: 5.65%	Month-to-month: 26.51%	20.85 pp
Satisfaction	Score 9: 7.48%	Score 1: 16.72%	9.24 pp
Late payments	0: 9.06%	3+: 17.16%	8.10 pp
Number of services	6: 7.37%	1: 11.04%	3.67 pp
Charge per service	Lowest decile: 8.61%	Highest decile: 11.27%	2.66 pp

A2. Variables with little standalone separation

Variable	Key result
Age group	9.90%-9.99% churn
Gender	9.82%-9.93% churn
Education	9.86%-10.00% churn
Annual income	\$58,488 retained vs \$58,545 churned mean; identical \$48,955 median
Days since last interaction	44.49 retained vs 44.55 churned mean; median 31 for both
Average monthly usage	38.64 GB retained vs 37.75 GB churned mean
Monthly charges	\$86.57 retained vs \$85.29 churned mean
Payment method	9.88%-9.95% churn
Paperless billing	9.99% No vs 9.90% Yes
Credit score	678.63 retained vs 678.52 churned mean; median 680 for both

Note: The full notebook contains all code, sample counts, distributions and intermediate outputs. This report intentionally presents a selected narrative suitable for portfolio review.